$M = 4 \times 10^{-31}\, M_P$, $\phi_* = 5\, M_P$: shifts relative to PRyMordial + ΛCDM
ΔY_p [%]
Δ(D/H) [%]
$\Delta N_{\rm eff}$ today
T_J at parking [GeV]
β
one-loop law: range over A = 0.05, 0.1, 0.2, 0.5, 1, 1.5, 2, 3
one-loop law, $A = 1$
one-loop law, $A = 3$ (largest A in the band)
one-loop law, $A = 5$ (dark radiation during BBN)
one-loop law, $A = 10$ (dark radiation during BBN)
fixed law, $\alpha = 0.1$
fixed law, $\alpha = 0.01$
±2σ of the measurement, about PRyMordial + ΛCDM
Planck 2018: $\Delta N_{\rm eff} < 0.30$ (95%, one-tailed, +lensing+BAO)
BBN, 2 MeV – 20 keV (indicative; parking row)
not at the wall before BBN ($T_{\rm deliver} < 10$ MeV; $\beta < 1.066$)
surfs the electroweak kick ($\beta > 3.000$–3.005)
analytic QCD surfing threshold $\beta = 1.1074$
no BBN result at any A (BBN temperature below PRyMordial's floor, or no history)
band above the axis